

A phase field model for cellular interfaces

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Abstract

Organoids are a thriving field at the intersection of nonlinear physics, biology, and practical applications. New ways of modeling such systems are required. A phase field model for the description of cellular membrane shape is presented. The cellular interface is described by a continuous, sigmoid-like set of functions obeying a dynamics based on minimizing surface area while keeping a fixed volume. The model was built to be suitable for numerical simulations and used to reproduce the dynamics of two cells susceptible to adhesion, showing how the ratio between the latter and surface tension influences the shapes of cells during their division and the complete pattern that emerges at the end. The work is based on the cutting-edge work by Makiko Nonomura. [1].

The study of the dynamics of cells' shapes is demonstrating itself to be extremely useful in many frontier fields, both in research, such as cleavage modeling to study embryo growth in embryology, and applied fields: the modeling of competition between surface tension and adhesion and their numerical and experimental simulation through organoids can help understand under which biochemical conditions a cluster of cancer cells detaches from the primary core and flows away into the bloodstream. Previous models lacked in computational scalability and parametrical simplicity, both issues solved by the phase field model. Here, a vector field of continuous function is adopted. Each entry of the vector represents a cell. Each cell's interface is described with a continuous function that assumes the value of one in the cell's interior, zero elsewhere, acting as an order parameter with a smooth, sharp transition between the two

The time evolution of such functions is recovered through a set of cell-type-dependent PDEs, coupled together only by the excluded-volume interaction term. This approach prevents the use of fluctuations to carry out simulations, which are well-suited to be scaled up to many cells via the introduction of well-contrived auxiliary functions of the interface functions. In this work, the first basic results are recovered; in particular, the interaction between cells and substrate is neglected, cells are considered to be of the same type, and possess a two-dimensional shape. Adhesion of two cells is explored through contour plots of their interface functions.

Model

The time evolution of the interfaces functions is derived by taking the functional derivative of a free-energy potential $E[u]$.

The energy potential is built as a sum of contributions of different energy actors, each of them representing one of the competing mechanisms determining a cell shape.

$$E[u] = E_{cell}[u] + E_{interactions}[u] \quad (1)$$

In the following each term will be explicitly written and analyzed.

$$E_{cell}[u] = \int_{\Omega} \left[\frac{D_0}{2} |\nabla u|^2 + W(u) \right] d\mathbf{r} + \frac{\alpha}{12} (V_0 - v(u))^2 \quad (2)$$

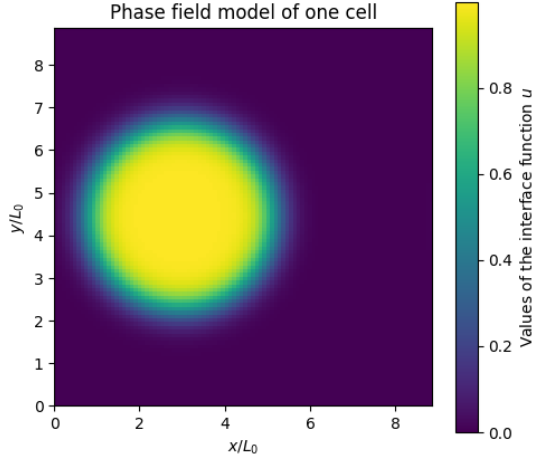
Since it is a functional that must be minimized, an increase in energy has the meaning of a discouraged behaviour and viceversa. The first term is a Landau-Ginzburg potential, the constant D should be interpreted as a diffusive constant, the first integrand represents the energy cost of a shape variation, the second one is a double-well potential which forces the function $u(x, y, t)$ to localize to one of the two meaningful states i.e. to keep the integrity of the cell. The shape of the potential is given by the function:

$$W(u) = \frac{1}{4} u^2 (1 - u)^2 + w_1 h(u) + w_o (1 - h(u)) \quad (3)$$

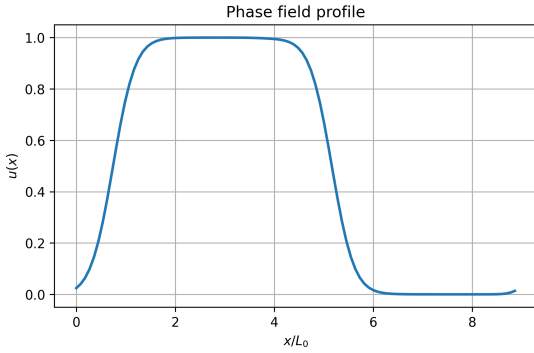
Where the coefficients w_o and w_1 control the depth of the wells, corresponding at the two states for the order parameter, and with:

$$h(u) = u^2 (3 - 2u) \quad (4)$$

The function $h(u)$ is defined such as $h(1) = 1$ and $h(0) = 0$ and it will be used in the computation of volume and cell's adhesion energy terms. Its cubic form ensures that the derivatives vanish at $u = 0$ and $u = 1$, providing numerical stability and a smooth transition across the diffuse interface (the cell membrane). the last one is a spring-like volume-constraint term. The parameter α controls the rate of volume's growth and the function as a whole simulates the incompressibility of the since it increases energy if cell's volume deviates from its target



(a) Phase field model: the interface function u is 1 inside the cell and 0 elsewhere. The thickness of the membrane is of order $O(\sqrt{D_0})$.



(b) The function u ensures a smooth transition between the cell interior and exterior thanks to a sigmoid-like profile.

Figure 1: Phase-field description of a single cell and its diffuse interface.

ones.

$$\begin{aligned}
 E_{\text{int}} = & \sum_{m=1}^N \int_{\Omega} \left[\frac{\beta}{12} h(u_m) \sum_{\substack{n=1 \\ n \neq m}}^N h(u_n) \right. \\
 & - \frac{\eta}{12} \nabla h(u_m) \cdot \sum_{\substack{n=1 \\ n \neq m}}^N \nabla h(u_n) \\
 & \left. + \frac{\gamma}{12} |\nabla h(u_m)|^2 \right] d\mathbf{r}. \quad (5)
 \end{aligned}$$

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The study of the dynamics of cell shapes is proving to be extremely useful in many frontier fields, both in research—such as cleavage modeling to study embryo growth in embryology—and in applied fields. The modeling of the competition between surface tension and adhesion, and their numerical and experimental simulation through organoids, can help understand the biochemical conditions under which a cluster of cancer cells detaches from the primary core and flows into the bloodstream.

Previous models lacked computational scalability and parametric simplicity, both issues solved by the phase field model. Here, a vector field of continuous functions is adopted. Each entry of the vector represents a cell. Each cell's interface is described by a continuous function that assumes the value of one in the cell's interior and zero elsewhere, acting as an order parameter with a smooth but sharp transition between the two.

The time evolution of such functions is recovered through a set of cell-type-dependent PDEs, coupled together only by the excluded-volume interaction term. This approach prevents the use of fluctuations to carry out simulations, which are well-suited to be scaled up to many cells via the introduction of well-contrived auxiliary functions of the interface functions. In this work, the first basic results are recovered; in particular, the interaction between cells and substrate is neglected, cells are considered to be of the same type, and possess a two-dimensional shape. Adhesion of two cells is explored through contour plots of their interface functions.

The term $\beta h(u_i)h(u_j)$ represents a repulsive force. Since $h(u) \approx 1$ inside the cell, this integral is non-zero only when two cells overlap. The constant $\beta > 0$ ensures that cells do not occupy the same space. The term $-\eta \nabla h(u_i) \cdot \nabla h(u_j)$ models the attractive force between membranes. The gradient $\nabla h(u)$ is non-zero only at the cell boundary. This term lowers the total energy when two cell membranes are in close contact. The last one resembles surface tension; if γ is raised, the cell is incentivized to keep a spherical, gentle boundary.

Finally, having written down the energy functionals for two cells and imposing $\tau \partial_t u = -\frac{\delta E}{\delta u}$ with $\tau > 0$, the dynamics are obtained through the system of equations displayed in (6). The latter equality is interpreted in the context of out-of-equilibrium thermodynamics: the time evolution of the order parameter of the system is guided by minimizing the total free energy functional dependent on that parameter.

$$\begin{aligned}
 \partial_t u_1 = & D\Delta u_1 + u_1(1 - u_1) \left[u_1 - \frac{1}{2} + \alpha(V_{tg} - V_1) \right. \\
 & \left. - \beta h(u_2) + \eta \Delta h(u_2) + \gamma \Delta h(u_1) \right] \\
 \partial_t u_2 = & D\Delta u_2 + u_2(1 - u_2) \left[u_2 - \frac{1}{2} + \alpha(V_{tg} - V_2) \right. \\
 & \left. - \beta h(u_1) + \eta \Delta h(u_1) + \gamma \Delta h(u_2) \right] \quad (6)
 \end{aligned}$$

Results

Figure 2 shows the contour plots at $u = 0.2$ for different adhesion strenghts. The snapshots are taken after 20000

time steps at $u = 0.2$. The units of the x-axis are rescaled with respect to the typical initial cell size $L_0 = \sqrt{\frac{V_{target}}{\pi}}$. A slight deformation of the cell membrane is observed; the small "elliptical intersection" can be adjusted by varying the parameter β .

Figure 2d shows the profile of the interface functions u along the line connecting the two cell centers. The function e_η represents an adhesion density and it is very sensitive to membrane overlaps, its definition is:

$$e_\eta := \sum_{m=1}^2 \frac{\eta}{6} \nabla h(u_m) \cdot \nabla h(u_{n \neq m}) \quad (7)$$

The results on cell division explores the effects of adhesion, diffusion and surface tension and they are summarized in figure 3. The results are in good agreement with Nomiura's original work.

Numerical Methods

The model is suitable to be explored by powerful numerical methods, but additional abstract variables must be solved in order to manage a high number of cells. As a first attempt, two cells can be modeled by keeping the functions defined above, each of them easily interpretable. As a first approach, a complex implicit Euler method was tried to discretize the dynamical equations in time and space, using a five-point discrete approximation for the Laplacian and a column vector whose entries were the points of the numerical grid, following a natural ordering for the function u . Periodic boundary conditions were applied directly to the matrix. This approach led to a nonlinear system of equations addressed by Newton's method. However, this strategy turned out to be over-complex, without leading to better results with respect to an explicit Euler scheme, which was therefore the final choice.

Cell division is implemented in the phase-field framework by monitoring the volume of each cell, whose spatial integral is approximated through a Riemann sum. Cell growth is modeled by a slow, linear increase of the target volume over time. When the actual cell volume exceeds a prescribed threshold, division is triggered numerically by splitting the phase field into two complementary daughter fields using a smooth sigmoidal mask along a fixed spatial direction. The mother cell is then removed and replaced by the two daughter cells, whose target volumes are reset. The sigmoid mask:

$$u_1(\mathbf{r}, t) = \frac{u_1(\mathbf{r}, t)}{2} \left[1 + \tanh \left(\frac{(\mathbf{r} - \mathbf{r}_c) \cdot \mathbf{n}_1}{\delta} \right) \right] \quad (8)$$

$$u_2(\mathbf{r}, t) = \frac{u_2(\mathbf{r}, t)}{2} \left[1 - \tanh \left(\frac{(\mathbf{r} - \mathbf{r}_c) \cdot \mathbf{n}_2}{\delta} \right) \right] \quad (9)$$

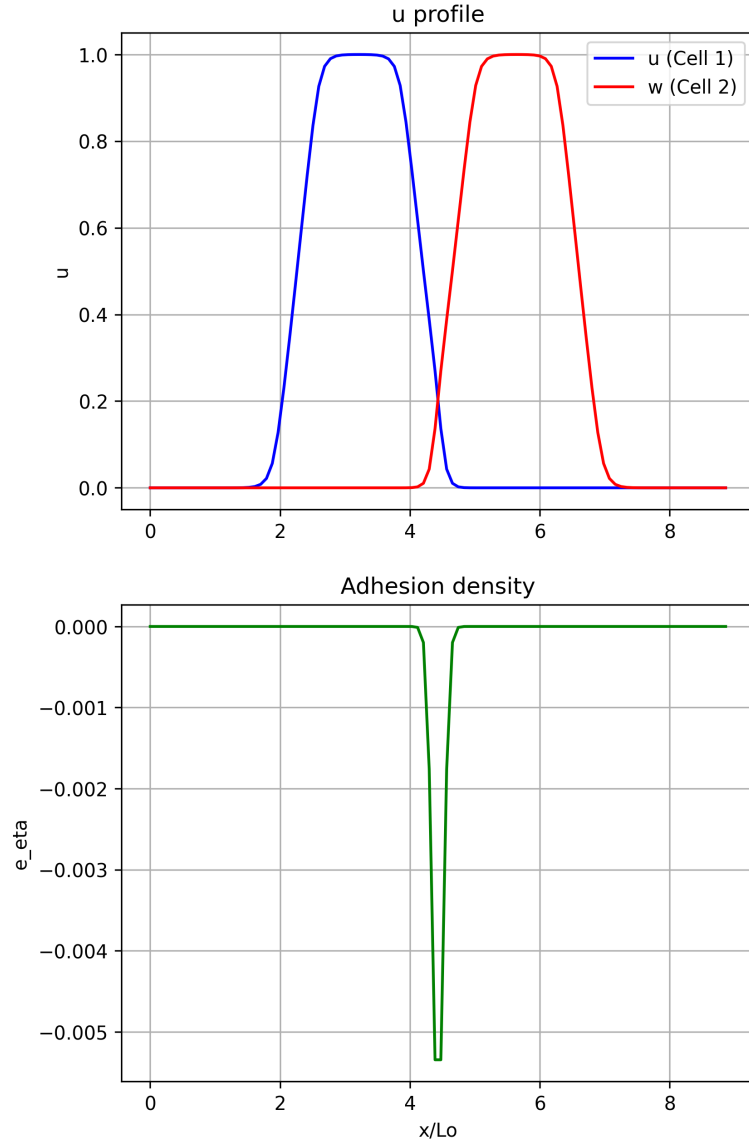
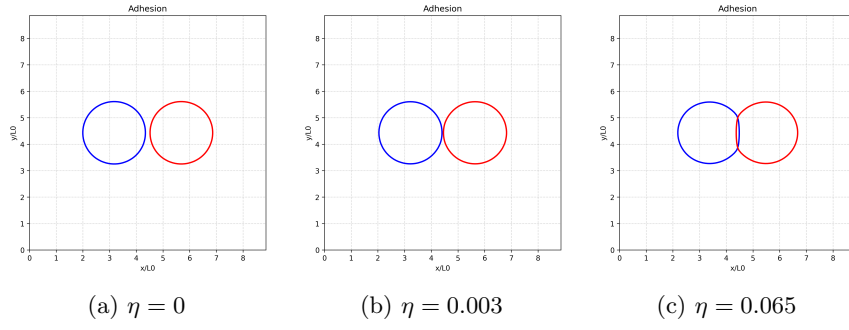
This approach offers the advantage of preserving the total volume through a straightforward division rule. Although the sigmoid mask enforces a specific division axis—which, in theory, should be stochastic or determined by the cell's morphology—significant differences in shape and pattern emergence remain observable across different parameter tunings. These observations could form the basis for a first extension of the model.

Conclusions

In this work, we reproduced and analyzed a phase-field model for a two-cell system, demonstrating how cell shape, volume conservation, adhesion, and surface tension collectively determine cellular morphology and interactions. The model naturally captures interface dynamics and excluded-volume and well reproduces the results of Nomiura's original work despite the simplifying decisions undertaken. Beyond, this framework opens the way to the study of collective phenomena such as cellular packing, cell sorting, and tissue-level morphology. To further refine the model and bring it closer to real-world cells and cellular environments, possible extensions include the introduction of heterogeneous cell types, explicit cell-substrate adhesion, and chemotactic responses under volume constraints and stochastic noise. Even beyond Nomiura's work, coupling phase-field dynamics with time-dependent volume constraints or growth laws could be proposed as further developments.

References

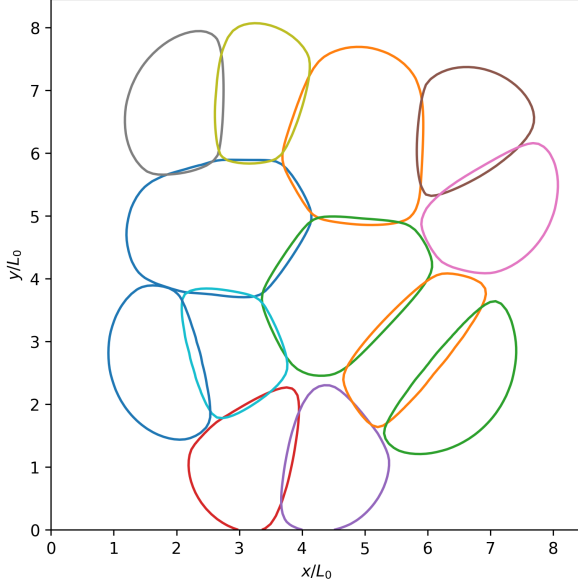
- [1] Makiko Nonomura. "Study on Multicellular Systems Using a Phase Field Model". In: *PLoS ONE* 7.4 (2012). PMID: 22539943; PMCID: PMC3335162, e33501. DOI: 10.1371/journal.pone.0033501. URL: <https://doi.org/10.1371/journal.pone.0033501>.



(d) Results for profiles of the functions u_1 , u_2 , and e_η along the line connecting the centers of the two cells.

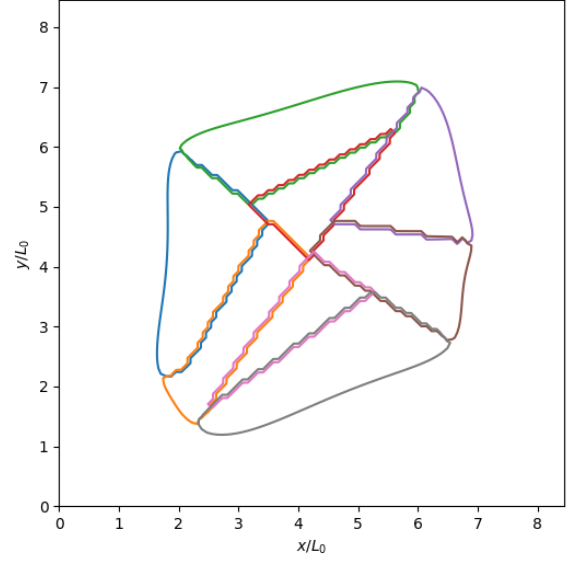
Figure 2: Overview of the cell interaction simulations: top panel shows the effect of different adhesion strengths on cell overlap, bottom panel shows the corresponding interface profiles along the line connecting the cell centers.

Step 8200 - $D = 0.0001$ - $\eta/\gamma = 0.20$ - Cell number: 13



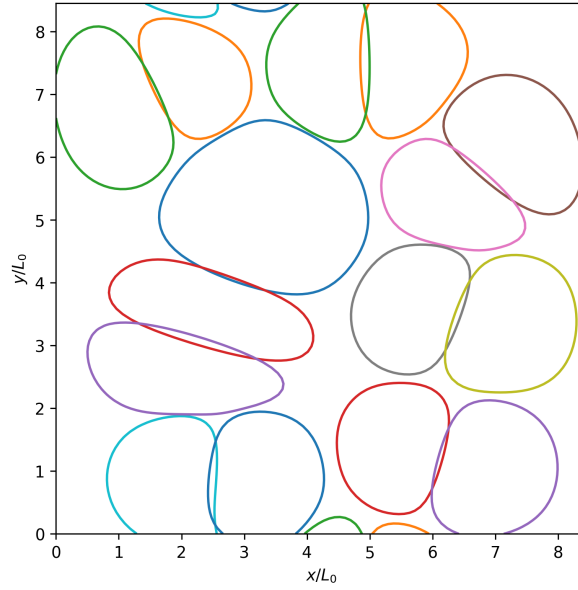
(a) Surface tension overcomes adhesion: cells show gentle, round borders, gaps are present in the grapes-shape structure, cells separates but never manage to occupy the whole volume available.

Step 5600 - $D = 0.0010$ - $\eta/\gamma = 1.20$ - Cell number: 8



(b) Adhesion overcomes surface tension: interfaces at the intersections between cells are zigzagged, jagged, cells don't separate at all and elongate; keeping the simulation running can lead to divergences.

Step 8800 - $D = 0.0100$ - $\eta/\gamma = 0.20$ - Cell number: 15



(c) Diffusion effects are enhanced with the same conditions for surface tension and adhesion: cells spread and occupied more volume, the final structure is less compact, cells have more time to grow without sensing adhesion effects

Figure 3: Comparison of cell division simulations under different parameter regimes. Top left: surface tension dominates; top middle: adhesion dominates; bottom: diffusion effects enhanced. Axes were normalized: a cell occupied a volume of, for example, 1.4 the original one means it has grown 40% its original size.